Block practice · File sizes, units and compression

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Pixels, resolution and colour depth

Kilobytes and kibibytes

Sound sampling: analogue to digital

Expressions for file size and transfer time

Compression: lossy or lossless?

OLevels.com SCHOOL

Practice workbook · Pixels, resolution and colour depth

Concepts of bitmap representation: what a pixel is, measures of resolution, and why more colour depth makes a bigger file.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-10

1. Question 3 · part b.ii · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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1. Question 3 · part b.ii · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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P 7 2 9 3 7 A 0 9 2 4

1. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 1 · part g.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.i only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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2. Question 1 · part g.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.i only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

.....

(f) Give the number of bits per character used by standard ASCII.

(1)

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

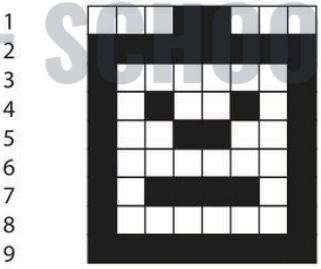


Figure 1

(i) State what is meant by the term **pixel**.

(1)

.....

(ii) Construct an expression to show the number of pixels in the image.

(1)

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2. Question 1 · part g.i · 1 mark · Working and answer

Working and answer

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3. Question 3 · part b.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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8

P 7 2 9 3 7 A 0 8 2 4

3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

4. Question 3 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

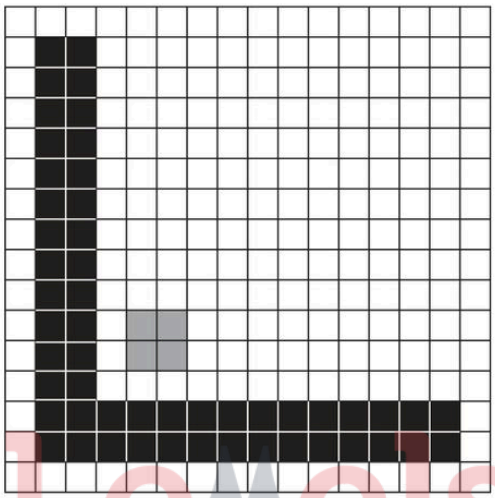


Figure 1

(i) State the smallest component of a bitmap image.

(1)

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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8

P 7 8 6 3 7 A 0 8 2 0

4. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Kilobytes and kibibytes

Kilobyte vs kibibyte and the other prefixes; converting a stated quantity between decimal and binary multiples.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-16

1. Question 2 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

P 7 2 9 3 7 A 0 5 2 4

1. Question 2 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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1. Question 2 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

(2)

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

(2)

(Total for Question 2 = 13 marks)

1. Question 2 · part e · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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14

P 7 8 6 3 7 A 0 1 4 2 0

3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

(c) Wearable technology, such as smartwatches, has become very popular with many groups of people including athletes.

Discuss how the use of wearable technology can be a benefit to health.

(6)

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(Total for Question 5 = 16 marks)

3. Question 5 · part b.iv · 1 mark · Working and answer

Working and answer

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Practice workbook · Sound sampling: analogue to digital

How analogue sound becomes digital: the sampling process, reading/plotting a sampled waveform and its axes, sampling frequency and the effect of changing it.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-13

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

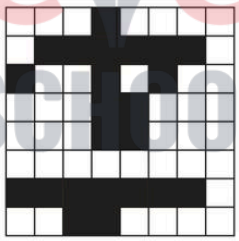


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

Figure 2 shows an overview of the process of recording a song and storing it as a digital sound file.

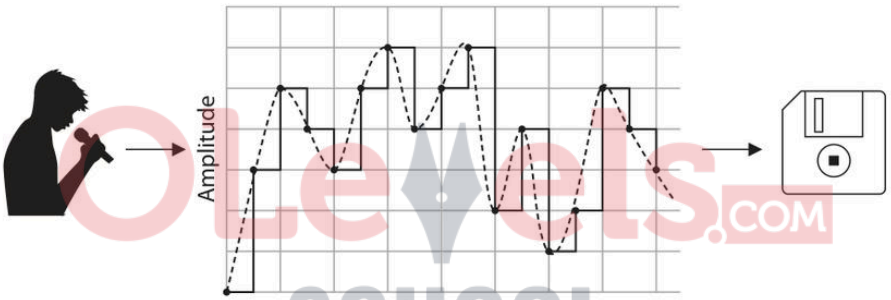


Figure 2

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

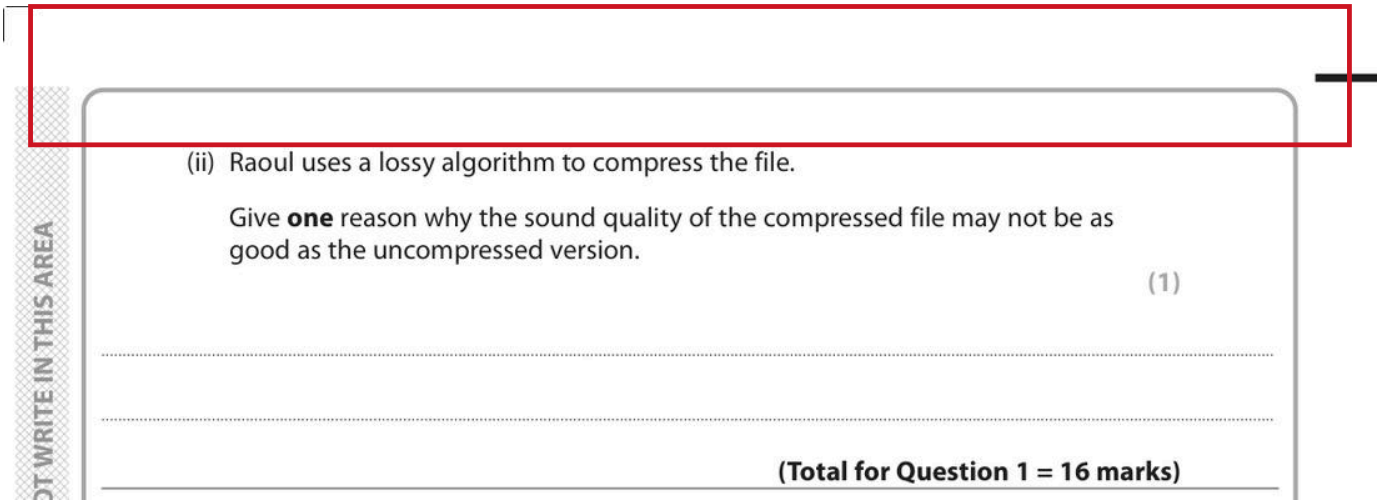
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P 6 9 2 9 7 A 0 4 2 0

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.



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(ii) Raoul uses a lossy algorithm to compress the file.

Give **one** reason why the sound quality of the compressed file may not be as good as the uncompressed version.

(1)

.....

.....

(Total for Question 1 = 16 marks)

Working and answer

OLevels.COM — SCHOOL —

2. Question 4 · part a.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

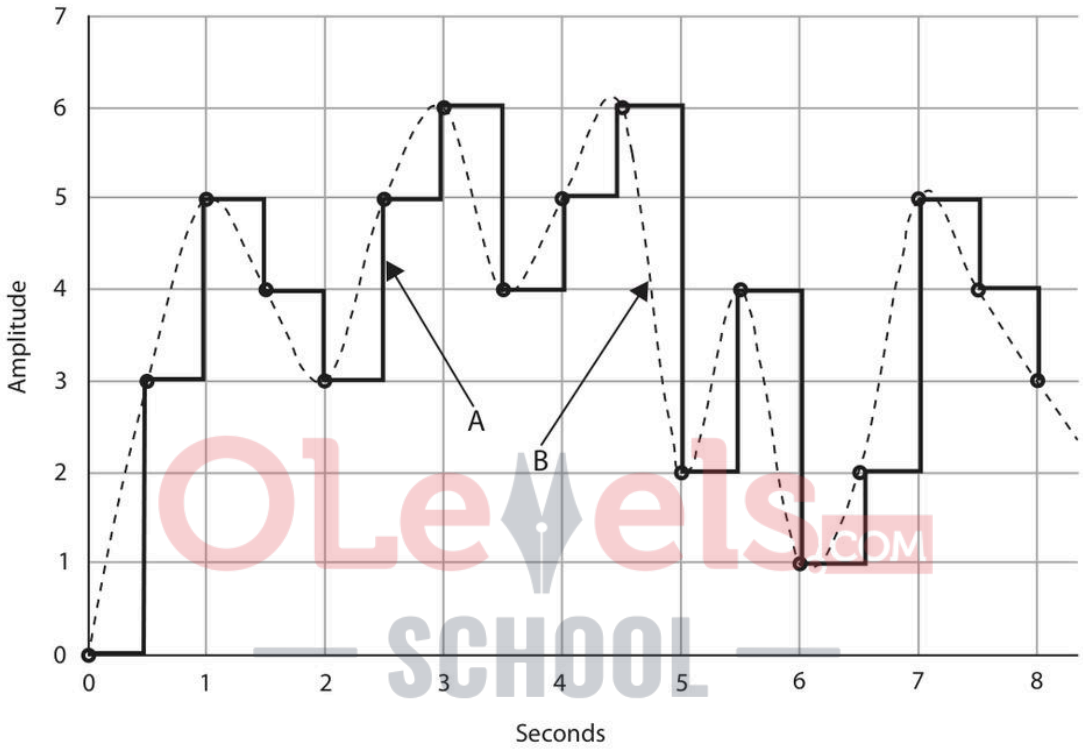


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

2. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.ii only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

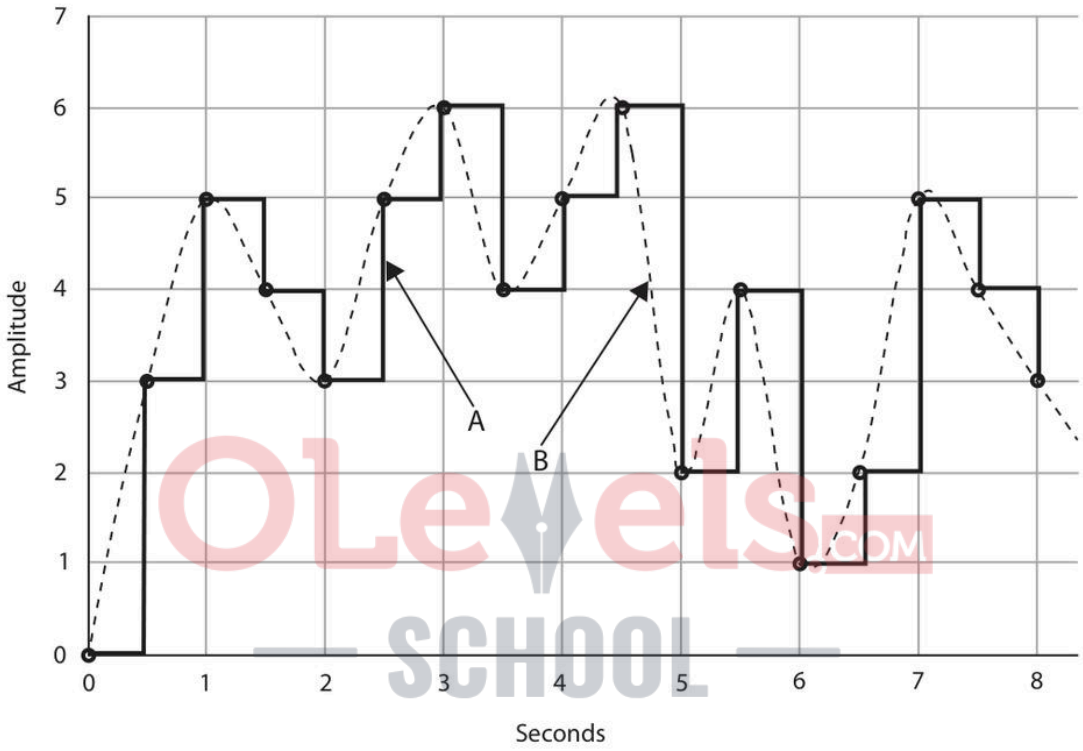


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(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

3. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iii only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

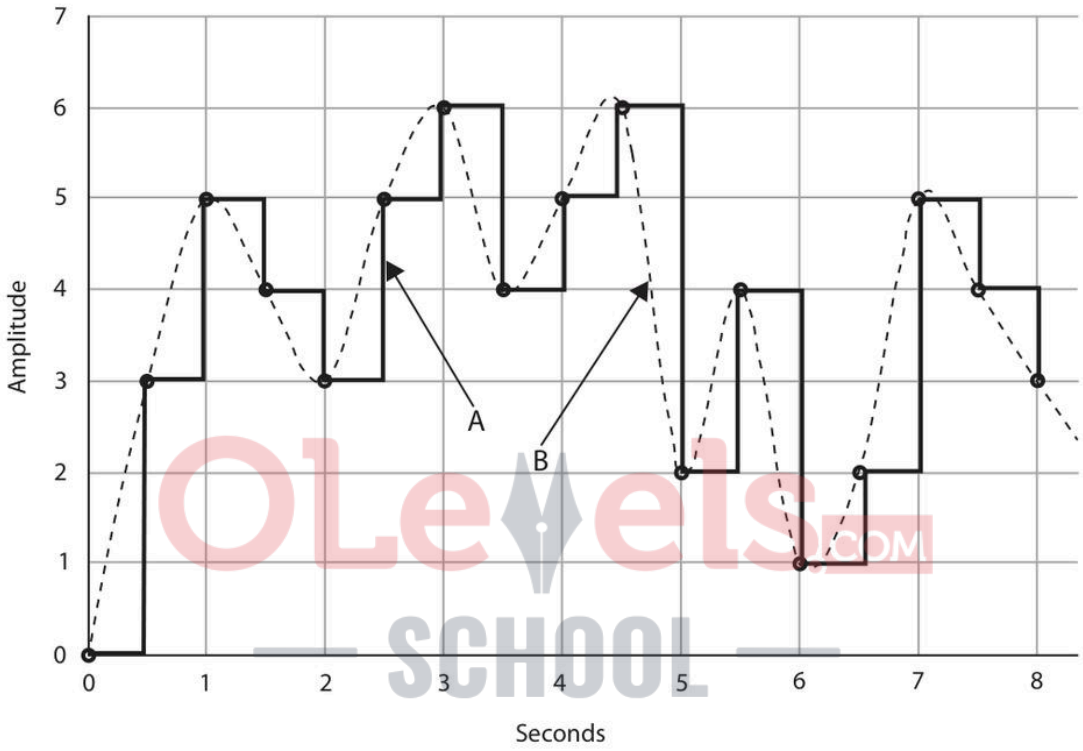


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

4. Question 4 · part a.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iii only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

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4. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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5. Question 4 · part a.iv · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iv only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

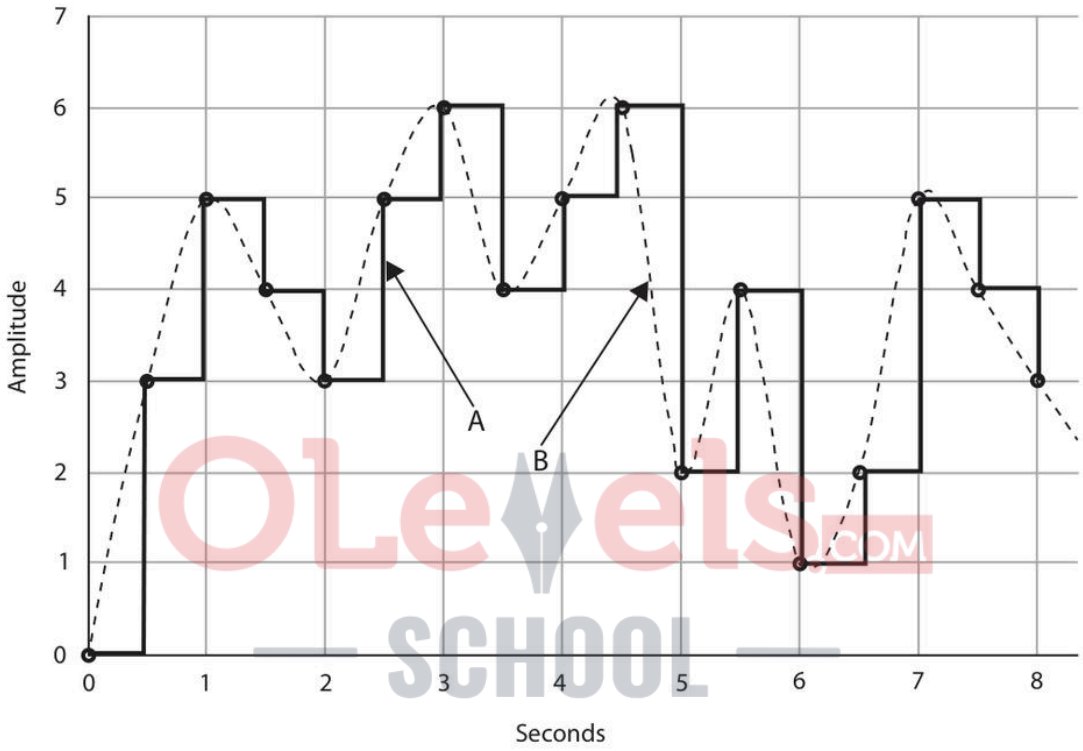


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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5. Question 4 · part a.iv · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iv only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

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5. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part a.i · 3 marks

2022 JUN · Paper 01

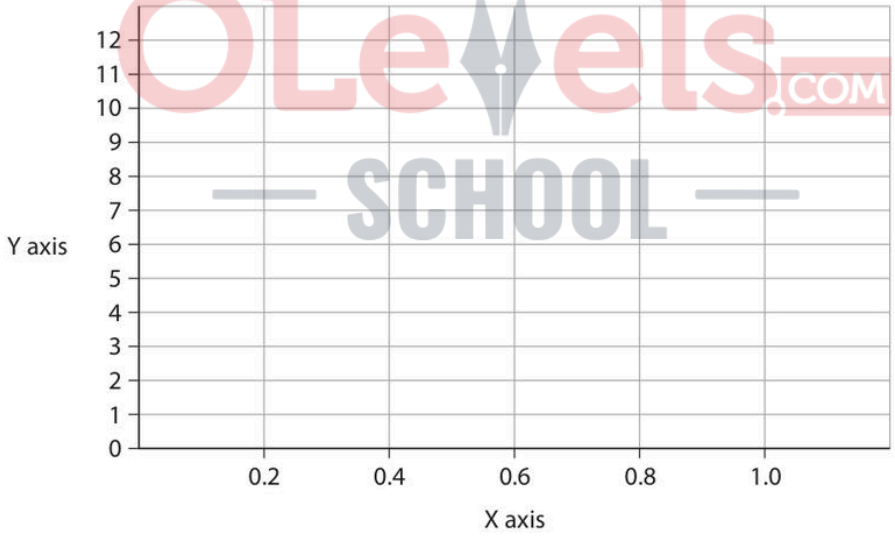
Further practice · Attempt part a.i only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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6. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

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7. Question 3 · part a.ii · 1 mark

2022 JUN · Paper 01

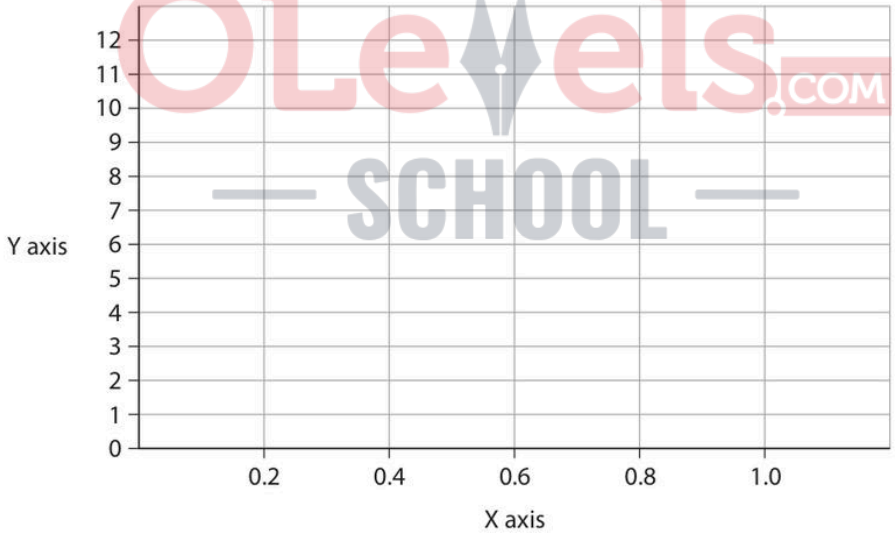
Further practice · Attempt part a.ii only.

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Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

.....

(iii) Give a suitable label for the Y axis.

(1)

.....

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P 7 2 5 3 8 A 0 8 2 0

7. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part a.iii · 1 mark

2022 JUN · Paper 01

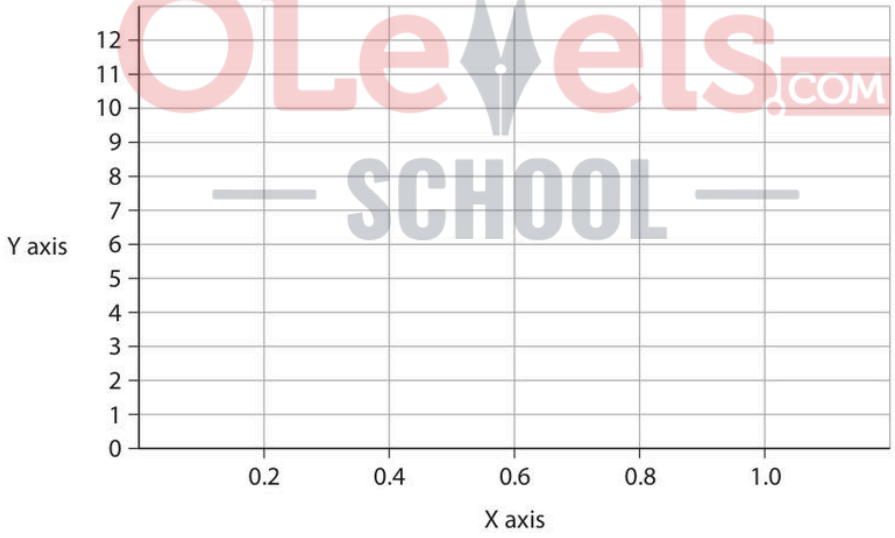
Further practice · Attempt part a.iii only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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8

P 7 2 5 3 8 A 0 8 2 0

8. Question 3 · part a.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.iii only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

.....

.....

.....

.....

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

8. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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Practice workbook · Expressions for file size and transfer time

Building (not evaluating) an expression for the size of an image, sound or text file, how many files fit in a store, or how long a file takes to send at a given bit rate, with all unit conversions.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-11

1. Question 2 · part d.ii · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

2 Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

(2)

(ii) Identify the number of characters that can be represented using standard ASCII.

(1)

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

(2)

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

(2)

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1. Question 2 · part d.ii · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(c) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	1	0	0
0	0	0	1	0	1	1	0

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

(2)

Colour	Binary pattern
Green	
Black	
White	
Red	
Blue	

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1. Question 2 · part d.ii · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(ii) Another image is 3579 pixels high and 6128 pixels wide.

The image is stored with a 32-bit colour depth.

The metadata for the image is 732 bytes.

Construct an expression to show how the file size, in **megabytes**, is calculated.

You do **not** need to do the calculation.

(4)

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(Total for Question 2 = 15 marks)

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1. Question 2 · part d.ii · 4 marks · Working and answer

Working and answer

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2. Question 4 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

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4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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2. Question 4 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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12

P 6 1 8 8 0 A 0 1 2 2 4

2. Question 4 · part b · 4 marks · Working and answer

Working and answer

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3. Question 4 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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3. Question 4 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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3. Question 4 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

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(d) A text file is stored on a hard disc.

The file holds information from one side of a sheet of paper.

The sheet of paper is represented as a grid, 80 columns wide and 66 rows long.

Each cell in the grid contains a single 2-byte Unicode character.

The file also contains 40 characters of metadata.

The hard disc allocates space in blocks of 1024 bytes.

Construct an expression to show the number of blocks required to store the file.

You do **not** need to do the calculation.

(4)

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(Total for Question 4 = 14 marks)

3. Question 4 · part d · 4 marks · Working and answer

Working and answer

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4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

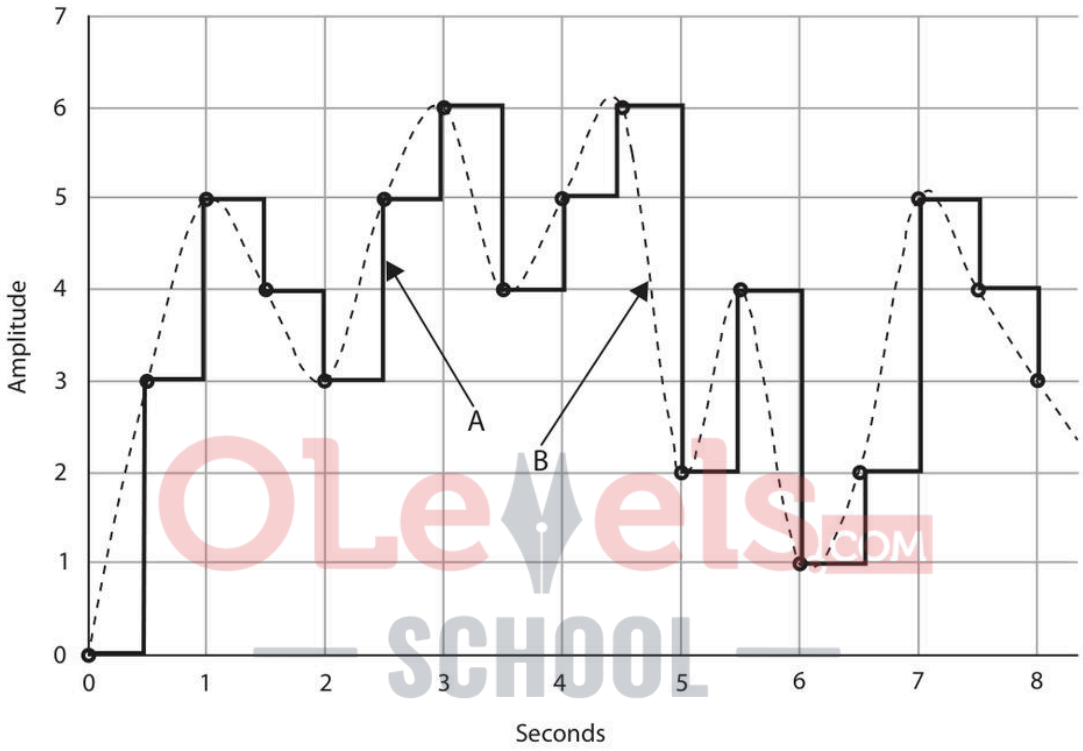


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

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(vi) Give the value in binary of the audio sample at the 4th second.

(1)

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

(4)

(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

(1)

(Total for Question 4 = 12 marks)

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4. Question 4 · part b · 4 marks · Working and answer

Working and answer

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5. Question 1 · part g.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.ii only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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5. Question 1 · part g.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.ii only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

(f) Give the number of bits per character used by standard ASCII.

(1)

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

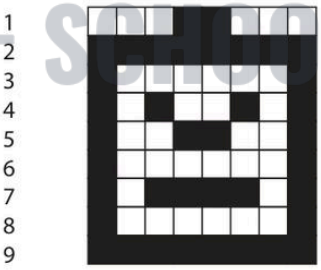


Figure 1

(i) State what is meant by the term **pixel**.

(1)

(ii) Construct an expression to show the number of pixels in the image.

(1)

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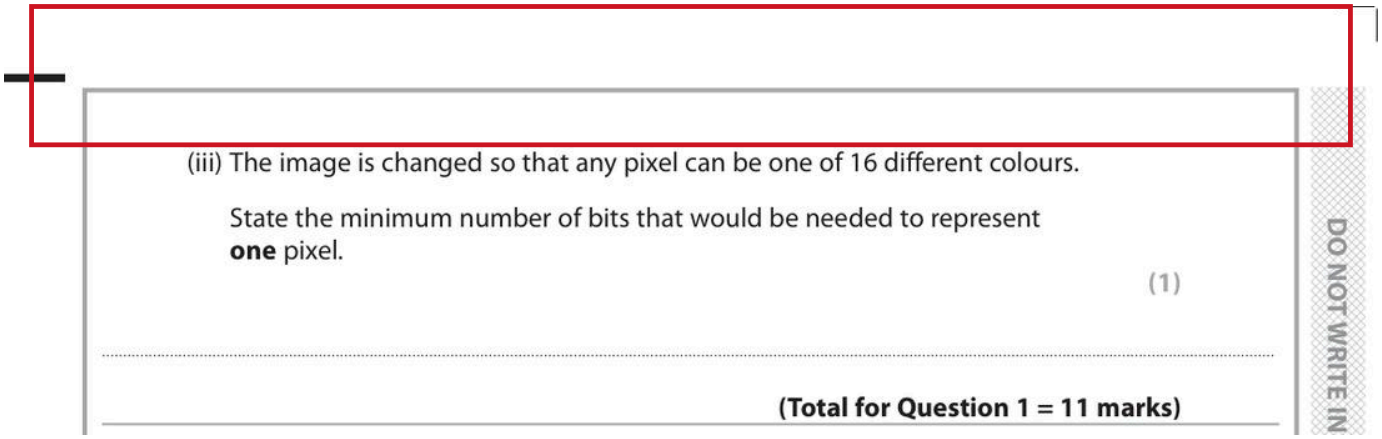
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5. Question 1 · part g.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.ii only.



(iii) The image is changed so that any pixel can be one of 16 different colours.

State the minimum number of bits that would be needed to represent **one** pixel.

(1)

.....

(Total for Question 1 = 11 marks)

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Working and answer

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6. Question 2 · part g · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part g only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

6. Question 2 · part g · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part g only.

(e) **Figure 2** shows a network topology.

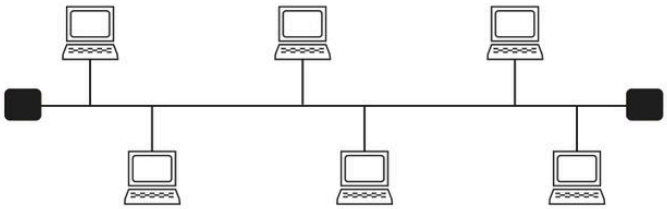


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1

2

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6

P 6 6 4 8 9 A 0 6 2 0

6. Question 2 · part g · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part g only.

(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

CCCWWWCCWWWWWWCCC

Give the result of compressing the string using a run-length encoding algorithm.

(1)

.....

.....

(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

(Total for Question 2 = 18 marks)

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6. Question 2 · part g · 4 marks · Working and answer

Working and answer

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7. Question 1 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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7. Question 1 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

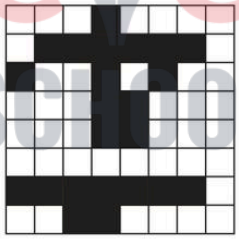


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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7. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 3 · part a.ii · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a.ii only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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8. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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9. Question 3 · part b.ii · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

9. Question 3 · part b.ii · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

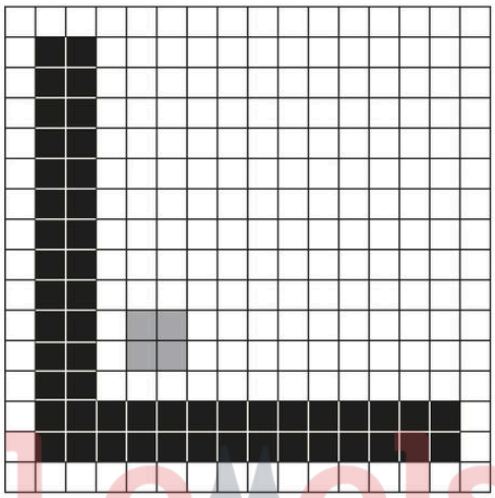


Figure 1

(i) State the smallest component of a bitmap image.

(1)

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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9. Question 3 · part b.ii · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

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(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

(2)

(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

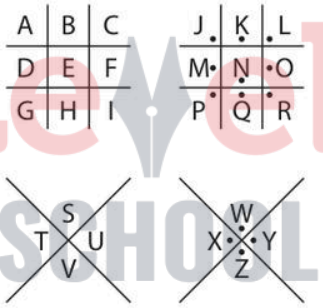
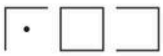


Figure 2

Complete the table to give the decrypted plain text.

(2)

Encrypted text	Plain text
	
	

(Total for Question 3 = 14 marks)

9. Question 3 · part b.ii · 4 marks · Working and answer

Working and answer

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Practice workbook · Compression: lossy or lossless?

Why files are compressed, drawbacks of compression, the meaning and examples of lossy and lossless, and choosing and justifying a method for a scenario.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-14

1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

Issue	Utility software
Her word processor cannot read all of a file because it is corrupt.	
She has a JPEG image and needs to send it to a publisher who can only read PNG image files.	
There has been a flood and files on a server have been damaged, but Anna needs to access her data.	
Her hard drive is running very slowly when she tries to access a file.	

(4)

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1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

.....

Justification

.....

.....

.....

.....

(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

(Total for Question 5 = 15 marks)

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1. Question 5 · part d · 3 marks · Working and answer

Working and answer

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2. Question 4 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

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4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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2. Question 4 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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2. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(d) A text file is stored on a hard disc.

The file holds information from one side of a sheet of paper.

The sheet of paper is represented as a grid, 80 columns wide and 66 rows long.

Each cell in the grid contains a single 2-byte Unicode character.

The file also contains 40 characters of metadata.

The hard disc allocates space in blocks of 1024 bytes.

Construct an expression to show the number of blocks required to store the file.

You do **not** need to do the calculation.

(4)

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(Total for Question 4 = 14 marks)

3. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

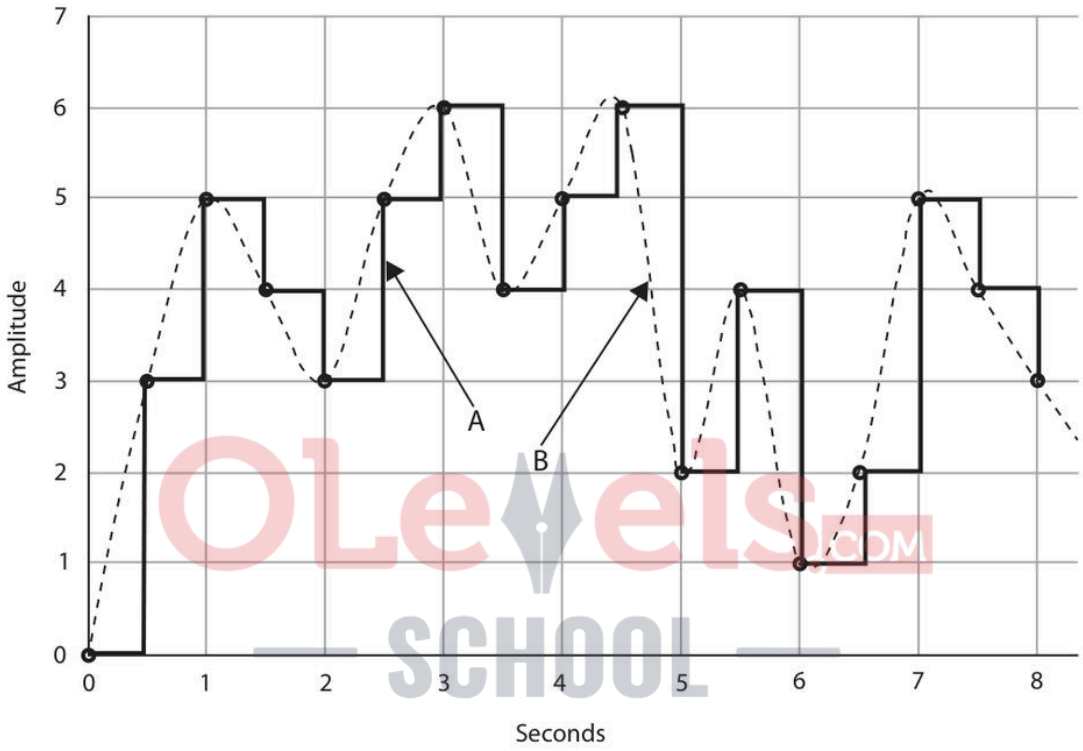


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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10

P 6 1 8 8 4 R R A 0 1 0 2 0

4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

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DO NOT WRITE IN THIS AREA

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

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(vi) Give the value in binary of the audio sample at the 4th second.

(1)

4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

(4)

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(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

(1)

(Total for Question 4 = 12 marks)

4. Question 4 · part c · 1 mark · Working and answer

Working and answer

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5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

(b) State the purpose of network protocols.

(1)

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

(e) **Figure 2** shows a network topology.

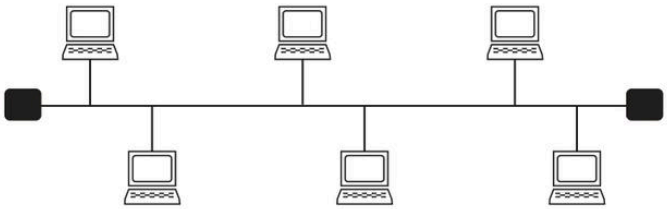


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1

2

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5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

DO NOT WRITE IN THIS AREA

(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

CCCWWWCCWWWWWWCCC

Give the result of compressing the string using a run-length encoding algorithm.

(1)

DO NOT WRITE IN THIS AREA

(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

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(Total for Question 2 = 18 marks)

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5. Question 2 · part f.i · 2 marks · Working and answer

Working and answer

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6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

(b) State the purpose of network protocols.

(1)

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

(1)

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

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6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

(e) **Figure 2** shows a network topology.

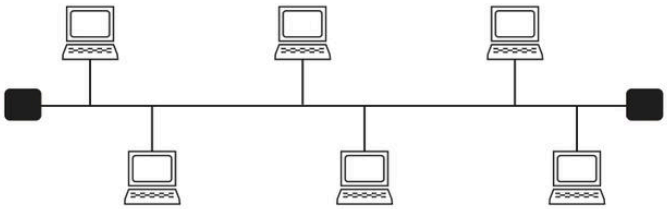


Figure 2

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(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1 .....

2 .....

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DO NOT WRITE IN THIS AREA

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6

P 6 6 4 8 9 A 0 6 2 0

6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

DO NOT WRITE IN THIS AREA

(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
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- ☒ **D** PNG

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.....

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- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

DO NOT WRITE IN THIS AREA

(Total for Question 2 = 18 marks)

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6. Question 2 · part f.ii · 1 mark · Working and answer

Working and answer

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7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

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DO NOT WRITE IN THIS AREA

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

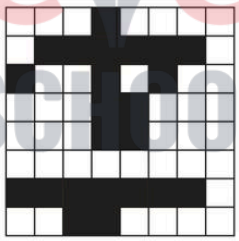


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

Figure 2 shows an overview of the process of recording a song and storing it as a digital sound file.

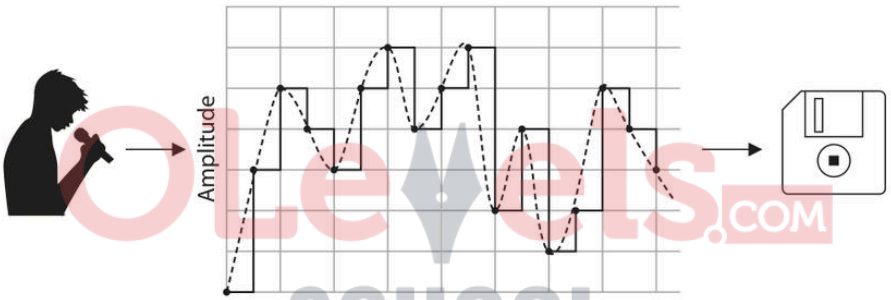


Figure 2

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

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4

P 6 9 2 9 7 A 0 4 2 0

7. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(2)

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8. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

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Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

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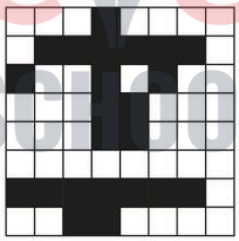


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Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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8. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

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Figure 2 shows an overview of the process of recording a song and storing it as a digital sound file.

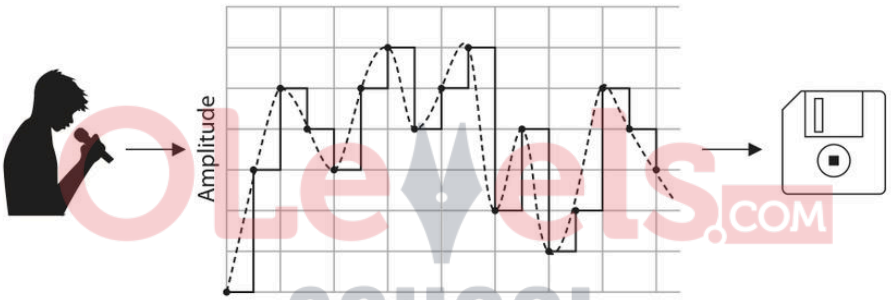


Figure 2

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

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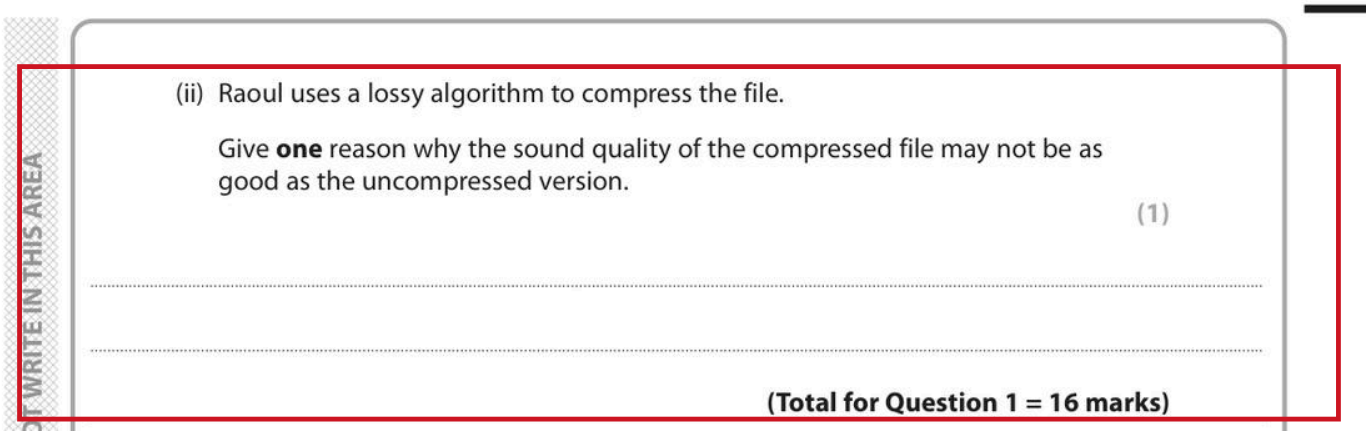
DO NOT WRITE IN THIS AREA

P 6 9 2 9 7 A 0 4 2 0

8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.



(ii) Raoul uses a lossy algorithm to compress the file.

Give **one** reason why the sound quality of the compressed file may not be as good as the uncompressed version.

(1)

.....

.....

(Total for Question 1 = 16 marks)

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Working and answer

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9. Question 3 · part b.i · 1 mark

2022 JUN · Paper 01

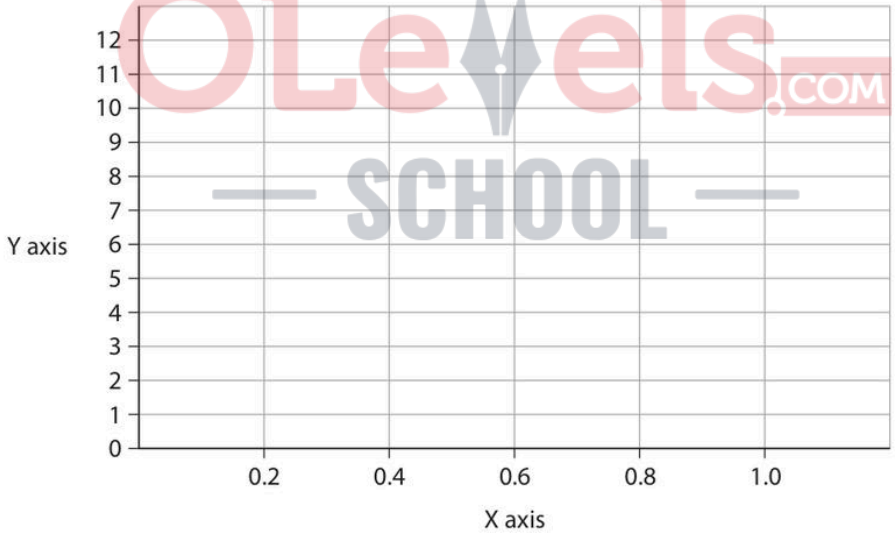
Further practice · Attempt part b.i only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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8

P 7 2 5 3 8 A 0 8 2 0

9. Question 3 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

.....

.....

.....

.....

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

9. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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